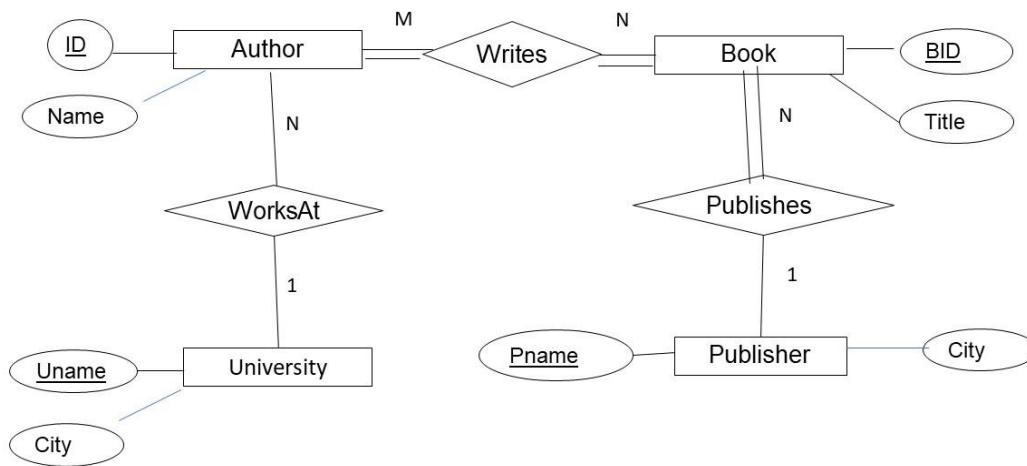


COMP 2411 Homework Assignment
(Deadline: Saturday, 15-Nov-2025, 11:59pm. Submit via Blackboard)

Students need to show their work in sufficient detail.

Q1: Map the ER diagram below to a relational schema diagram. You should use the approach presented in the lectures (which was aimed at minimizing the number of relations and the number of NULLs).



Assignment

Q2.

Suppose we store on disk a relation r with 4,000,000 records. K is a candidate key for r . The record size is 50 bytes, the block size is 500 bytes, each K value is 10 bytes long, and each block address is 7 bytes long.

2.1 For the unspanned block organization, calculate the minimal number of block I/Os necessary to find a record for a given K value, using a primary index on K . Of course, r is ordered on K .

2.2 For the unspanned block organization, calculate the minimal number of block I/Os necessary to find a record for a given K value, using a secondary index on K . Here, r is not ordered on K .

Q3.

Using the approach discussed in class, insert, in the order given, (records with) the following search key values into an initially empty B+tree: 4,6,7,3,9,2,5. Assume each leaf node can hold at most two data pointers, and each internal node can hold at most three tree pointers.

Q4. Let $R(A,B,C,D, E,F,G)$, and $F = \{ AB \rightarrow C, A \rightarrow D, BD \rightarrow C, D \rightarrow BG, AE \rightarrow F \}$

4.1. Determine the candidate keys of R.

4.2. Determine the highest normal form of R.

4.3. If R is not in 3NF, then decompose R to 3NF, and then to BCNF. If R is not in BCNF but in 3NF, then decompose R to BCNF.

For the next two questions (Q5 and Q6), consider the following three relations:

COURSE(CName, Area)
STUDENT(SID, Name, Gender)
PASSED(SID, CName, Year, Semester)

The underlined attributes form the primary keys. The COURSE relation stores the name and area of all courses in the CS Department. Example areas are “data” and “AI”. The STUDENT relation stores the SID, name and gender of students. The PASSED relation stores facts about which student passed which course in what year and semester. Consider the following queries:

B. Find the names of students who did not pass any course that Jim Luck passed. (“Did not pass any” means “did not pass even one.”)

C. Find the names of students who passed one or more courses with the “data” area and who also passed one or more courses with the “AI” Area.

D. Find the names of students who passed every course that Jim Luck passed.

E. For each student and year, find the total number of courses the student passed in that year.

Q5: Implement B and C above in relational algebra. You can reuse relational algebra expressions through intermediate relations, using meaningful intermediate relation names. Natural and concise solutions are preferred. Long and convoluted solutions will be penalized.

Q6: Implement B, D, E above in SQL. Natural and concise solutions are preferred. Long and convoluted solutions will be penalized.